The residual RMS bound, revised: the envelope-witness chain

figures `fig_bound_final_sim`, `fig_bound_final_hw`; scripts in the Reproduction section

Symbols

$\tau,\ \tau_{\text{end}}$	time since onset; window end	x	$= C_2\tau_{\text{end}}$, dimensionless window
y_i	measured rate at τ_i	N_s	number of samples in the window
C_1, C_2	nominal amplitude $K\dot{M}$ and exponent	$K,\ \dot{M}$	gain $1/Wz_{CoM}$; ramp rate
$W,\ z_{CoM},\ J_P$	weight, CoM height, pivot inertia	Wz_{CoM}	$= J_P C_2^2$ (the identity)
ω_{nom}	nominal rate $C_1(\cosh C_2\tau - 1)$	t_{on}	realised onset time
$e_\omega(\tau)$	true rate minus the nominal	$b,\ n_i$	gyro bias; noise sample
$\hat{h}(\cdot;\theta)$	fitted family member	θ	(C_1^*, C_2^*, t^*, C), all free
r_i	residual $y_i - \hat{h}(\tau_i;\hat{\theta})$	$E(\tau)$	envelope $\bar{\rho}KC_2\sinh(C_2\tau)$
$\rho(s)$	deviation forcing (small-angle + GE)	$\bar{\rho}$	window bound on $\lvert\rho\rvert$, eq. (12)
$B(x)$	$\frac{1}{4}\sinh 2x - \frac{x}{2}\ \left(= \int_0^x \sinh^2 u\, du\right)$	$\varphi_{\max}$	design tilt cap (5° sim, 8° hw)
$l_{\text{arm}},\ \beta_M$	pivot arm; GE coefficient	ΔM_{win}	in-window moment sweep $\dot{M}x/C_2$
n_{hi}	residual content above 5 Hz	κ_b	quiet-segment band ratio (1.31)
σ_n	declared sim gyro std, 2.450×10^{-4} rad/s		

1 The claim

For every run of both campaigns, with every quantity on the right computable before the experiment except the hardware vibration amplitude n_{hi} (read from the run's own out-of-band content),

$$\boxed{\mathrm{RMS}(r)\ \le\ \mathrm{RMS}(e_\omega) + \mathrm{RMS}(n)\ \le\ \bar{\rho}KC_2\sqrt{\tfrac{B(x)}{x}}\ +\ N.} \tag{1}$$

RMS(r) is the full-window RMS of the deployed PNLS fit's residual — the fit exactly as it runs in the pipeline, both shape constants and the onset free, no calibration anywhere. N is the noise term of Sec. 6: negligible ($\sigma_n = 0.014°$/s) in simulation, the measured vibration term $n_{\text{hi}}\sqrt{1+\kappa_b^2}$ on hardware. The two inequalities have entirely different characters and are proved separately: the first is an optimisation triviality (Sec. 3), the second is the physics (Sec. 4). Validation: 307/308 simulation runs and 140/140 hardware runs inside, the single exceedance located and explained (Sec. 7).

2 The measurement and the family

In the excitation window the record is

$$y_i\ =\ \omega_{\text{nom}}(\tau_i) + e_\omega(\tau_i) + b + n_i, \qquad \omega_{\text{nom}}(\tau) = \begin{cases} C_1\big(\cosh(C_2\tau) - 1\big), & \tau \ge 0, \\ 0, & \tau < 0, \end{cases} \tag{2}$$

with τ counted from the *realised* onset t_{on}: the instant the restoring moment actually reverses at the vehicle, wherever the actuation chain put it relative to the commanded ramp clock. The deviation $e_\omega = y - b - n - \omega_{\text{nom}}$ collects everything the small-angle model omits — trigonometric remainders, ground effect — and vanishes identically before the onset.

The estimator fits the four-parameter family

$$\hat{h}(\tau;\theta)\ =\ C_1^*\big(\cosh(C_2^*(\tau - t^*)) - 1\big)\,\mathbf{1}[\tau \ge t^*]\ +\ C, \qquad \theta = (C_1^*,\ C_2^*,\ t^*,\ C), \tag{3}$$

by least squares over the full window, $\hat{\theta} = \arg\min_\theta \sum_i \big(y_i - \hat{h}(\tau_i;\theta)\big)^2$, and $r_i = y_i - \hat{h}(\tau_i;\hat{\theta})$.

3 The first inequality: why the bound lands on e_ω

The point of this section is one sentence: *the nominal solution is itself a member of the fitted family, so the fit cannot lose to it.* Take the particular parameter vector

$$\theta_0 \;=\; (C_1,\; C_2,\; t_{\text{on}},\; b) \qquad\Longrightarrow\qquad \hat{h}(\tau;\theta_0) \;=\; \omega_{\text{nom}}(\tau) + b. \tag{4}$$

Nothing in (3) forbids this choice: the amplitude slot takes C_1, the exponent slot C_2, the onset slot the realised t_{on}, the baseline slot the bias. θ_0 is called the *witness*: not the minimiser, not a good fit, merely one admissible competitor whose residual can be computed exactly. A minimum is bounded by the value at any feasible point, so

$$\sum_i r_i^2 \;=\; \min_\theta \sum_i \big(y_i - \hat{h}(\tau_i;\theta)\big)^2 \;\le\; \sum_i \big(y_i - \hat{h}(\tau_i;\theta_0)\big)^2 \;=\; \sum_i \big(e_\omega(\tau_i) + n_i\big)^2, \tag{5}$$

where the last equality is (2) minus (4). Dividing by N_s and taking the square root, then splitting by the triangle inequality in the sample ℓ^2 norm,

$$\text{RMS}(r) \;\le\; \text{RMS}(e_\omega + n) \;\le\; \text{RMS}(e_\omega) + \text{RMS}(n). \tag{6}$$

That is the whole proof. Four remarks fix its scope.

(i) What is and is not claimed about the minimiser. (5) does not say the fit finds the nominal, or anything close to it: the minimiser wanders wherever the data pulls it (the amplitude–onset ridge of the earlier analysis is real). The inequality runs one way only — the fit does *at least as well* as the nominal — and one way is all a residual *upper* bound needs. The measured residuals sitting far below the cap (median used fraction 0.32–0.33 on both campaigns) is the minimiser exploiting freedoms the witness does not use. Geometrically: the measured curve lies between the nominal and the nominal-plus-envelope, and the fitted curve, free to move, lands at least as close to the measurement as the lower edge of that corridor — so the fit's residual is capped by the corridor's width, which is exactly $\text{RMS}(E)$.

(ii) No admissibility condition. The earlier draft's witness was the nominal *plus* a $\beta\sinh$ correction, folded back into the family through a time shift — legitimate only while $|\beta| < C_1$, which dragged the dimensionless ratio $u = \bar{\rho}C_2/\dot{M}$ and a lower ramp-rate limit into the certificate. The witness (4) is the nominal itself, a family member by construction, unconditionally. The $u < 1$ discussion survives only where it always belonged: whether a *fitted* branch has a physical onset to read off, not whether the residual is capped.

(iii) The onset freedom absorbs pure delay. Every transport and actuation lag between the commanded ramp and the vehicle shifts t_{on}, and nothing else: the tip-over dynamics start when the moment balance reverses at the vehicle, and carry no memory of the command clock. Because t^* is free in (3), the witness uses the realised onset, and e_ω — defined *against the realised-onset nominal* — obeys the deviation dynamics of Sec. 4 regardless of the delay's size. This is why no latency channel appears anywhere in (1): a pure delay is invisible to the chain. (A delay that *distorts* the ramp within the window is not pure and lands in ρ; the measured achieved-moment ramps are straight to 3%, and the antisymmetric part cancels in the direction-pair half-sum.)

(iv) Full window against post-onset window. $\text{RMS}(r)$ is reported over the full window (pre-onset segment included), while the envelope of Sec. 4 lives on the post-onset segment. The directions compose correctly: $e_\omega \equiv 0$ and $\hat{h}(\cdot;\theta_0) = b$ before the onset, so the pre-onset segment contributes only noise to the witness's residual, and

$$\text{RMS}_{\text{full}}(e_\omega) = \sqrt{\tfrac{T_{\text{post}}}{T_{\text{full}}}}\;\text{RMS}_{\text{post}}(e_\omega) \;\le\; \text{RMS}_{\text{post}}(e_\omega), \tag{7}$$

so bounding the post-onset RMS bounds the reported quantity a fortiori.

(v) Why no Δ_{pre} term. The manuscript's (108) charges the mean-square cost of a witness branch that starts *before* the true onset ($\delta t_c < 0$) — over $[\delta t_c, 0]$ that branch already draws a cosh curve while the record still sits at its baseline, and the mismatch must be paid. That cost belongs to the *shifted* witness of the earlier construction, not to the bound as such: the witness (4) keeps the realised onset, so before it both the witness and the record are at the baseline, the mismatch is noise alone, and $\Delta_{\text{pre}} \equiv 0$ identically. Dropping the shifted witness removes the one term that grew with the ramp rate ($\Delta_{\text{pre}} \propto C_1 = K\dot{M}$) — the term that pushed the earlier cap upward in rate while the measured residuals move the other way.

4 The second inequality: the envelope and its window RMS

The deviation obeys the manuscript's forced tip-over equation ((89) there): with e_φ the tilt deviation, $e_\omega = \dot{e}_\varphi$,

$$J_P\,\ddot{e}_\varphi \;=\; Wz_{CoM}\,e_\varphi \;+\; \rho(\tau), \qquad e_\varphi(0)=\dot{e}_\varphi(0)=0, \qquad |\rho(s)|\le\bar{\rho}\;\text{ on } [0,\tau_{\text{end}}]. \tag{8}$$

Duhamel with the hyperbolic kernel and one differentiation gives

$$e_\omega(\tau) \;=\; \frac{1}{J_P}\int_0^\tau \cosh\big(C_2(\tau-s)\big)\,\rho(s)\,ds; \tag{9}$$

bounding ρ by $\bar{\rho}$ and integrating the kernel,

$$|e_\omega(\tau)| \;\le\; \frac{\bar{\rho}}{J_P C_2}\,\sinh(C_2\tau) \;=\; \bar{\rho}KC_2\sinh(C_2\tau) \;\triangleq\; E(\tau), \tag{10}$$

the manuscript's envelope (90); the identity $KC_2 = 1/(J_PC_2)$ follows from $K = 1/Wz_{CoM}$ and $Wz_{CoM} = J_PC_2^2$. A pointwise bound gives an RMS bound by integration: with $u = C_2\tau$ and $x = C_2\tau_{\text{end}}$,

$$\text{RMS}_{\text{post}}(e_\omega) \;\le\; \bar{\rho}KC_2\sqrt{\frac{1}{x}\int_0^x \sinh^2 u\,du} \;=\; \bar{\rho}KC_2\sqrt{\frac{B(x)}{x}}, \qquad B(x) = \frac{\sinh 2x}{4} - \frac{x}{2}, \tag{11}$$

B by the double-angle identity $\sinh^2 u = \frac{1}{2}(\cosh 2u - 1)$. The factor $\sqrt{B(x)/x}$ is simply the window RMS of the shape $\sinh(C_2\tau)$: $\approx x/\sqrt{3}$ for small windows, $\approx e^x/(2\sqrt{2x})$ for long ones — the price of the instability's growing mode, paid honestly rather than absorbed by a fitted term.

5 The forcing bound $\bar{\rho}$ and the design tilt cap

The channels are the manuscript's (91), window-averaged with its reduction factors $R_\varphi = 1/7$ and $R_{GE} = 1/5$:

$$\bar{\rho} \;=\; R_\varphi\cdot\tfrac{1}{2}\,Wl_{\text{arm}}\,\varphi_{\max}^2 \;+\; R_{GE}\cdot|\beta_M|\,\Delta M_{\text{win}}\,\varphi_{\max}, \tag{12}$$

with the window quantities set a priori per ramp rate — the nominal tilt reaches the cap when

$$\sinh x - x \;=\; \frac{\varphi_{\max}\,Wz_{CoM}\,C_2}{\dot{M}}, \qquad \Delta M_{\text{win}} = \frac{\dot{M}\,x}{C_2}, \tag{13}$$

solved once per rate before any run. The simulator models no ground effect, so its $\bar{\rho}$ carries the first term alone, with the per-case arm $l_{\text{arm}} = l_p + p_{\text{off}}$, $z_{CoM} = 0.272$ m, and $J_P = J^{\text{CAD}} + m(z_{CoM}^2 + l_p^2)$; hardware uses the design box of Sec. VI-D ($z_{CoM} = 0.30$ m, arms 0.160/0.130 m, $|\beta_M| = 0.0345$).

$\varphi_{\max}$ is not a tuning knob: (10) holds on the window only while the trajectory stays inside the box on which (12) bounds ρ, so *the cap must dominate the realised in-window tilt*, and its value is set by each campaign's own geometry. In simulation the gate stops the window near 4° realised tilt: $\varphi_{\max} = 5°$. On hardware the slightly uneven ground of the test site lets the in-window tilt rise reach 7.66° (median 4.82°, quaternion-measured across all 140 runs): $\varphi_{\max} = 8°$. Every appearance of $\varphi_{\max}$ in (12)–(13) is monotone, so a cap that dominates the realised tilt only over-covers.

6 The noise term N

RMS(n) in (6) is whatever the sensor adds that the physics does not produce.

Simulation. The declared sensor model: additive Gaussian rate noise of standard deviation $\sigma_n = 2.450\times10^{-4}$ rad/s $= 0.014°$/s. This sits an order of magnitude below the smallest envelope value on the campaign, so the validation figure carries the envelope alone; including σ_n changes no count.

Hardware. The vehicle vibrates under its running rotors, and no pre-experiment model of that channel exists, so its amplitude is the one measured input of the chain: per run, n_{hi} is the RMS of the residual's content above 5 Hz — a band the family cannot occupy, since its fastest structural scale is $C_2/2\pi \approx 0.8$ Hz — extended to the full band by the campaign constant $\kappa_b = 1.31$, the below-5-Hz to above-5-Hz vibration ratio measured on quiet segments outside every excitation window:

$$N \;=\; n_{\text{hi}}\sqrt{1+\kappa_b^2}. \tag{14}$$

7 Validation, both campaigns

Per run: the deployed free fit's full-window residual RMS against the cap of (1). No calibration enters on either side.

campaign	runs	inside	worst used fraction	median	p90
simulation ($\varphi_{\max} = 5°$, envelope alone)	308	307	1.68	0.33	0.62
hardware ($\varphi_{\max} = 8°$, envelope + (14))	140	140	0.83	0.32	0.47

The single exceedance is S11/M_y at the fastest ramp (1.20 N m/s, used fraction 1.68): the injected offset $(38, 14)$ mm sits beyond the design rectangle whose arm the $\bar{\rho}$ of (12) assumes, so this run operates outside the box the bound is issued for — the certificate fails exactly where its own premise does, and nowhere else. Every hardware run is inside, the worst using 0.83 of its cap: whatever unmodelled disturbances the outdoor vehicle meets — gusts, contact irregularities at the pivot, channels (12) and (14) reserve no term for — they stay within the margin the caps carry.

The two campaigns validate *one* construction: the same chain (1), the same witness, the same envelope, the same design-box $\bar{\rho}$. The panels differ in exactly the channels the platforms differ in — the simulator declares its noise and models no ground effect, so its cap is the envelope alone; the vehicle adds ground effect (inside $\bar{\rho}$) and vibration (the term (14)). Figures `fig_bound_final_sim` and `fig_bound_final_hw` show every run.

8 What remains of the shaped decomposition

The earlier draft of this document ran the bound through the decomposition $e_\omega = \beta \sinh(C_2\tau) + \delta e_\omega$ with $\beta = e_\omega(\tau_{\text{end}})/\sinh(x)$ (manuscript (95)–(97)), capping δe_ω by the Dirichlet comparison constants $1/12$ and $1/(2e)$ (manuscript (107)). Measured directly — pinned-theory fit, β from the definition, δe_ω pointwise — that cap does not cover: on the simulation campaign 0 of 301 runs sit inside it (realised δe_ω RMS 2–3× the cap), and on hardware only the below-5-Hz band clears it, marginally (101/139). The decomposition is an identity and survives as one; what fails is the premise that the remainder after the end-matched sinh subtraction is dominated by the in-family forcing response — the subtraction is exact only at the two window ends, and the out-of-family content between them is amplified by the same sinh growth the envelope charges for.

Its residual role is therefore qualitative: it explains *why the measured residuals are flat in the ramp rate* (the fit's onset freedom absorbs the growing mode's dominant multiple, which is the tangent direction of the time shift), while the quantitative certificate is (1). For the manuscript this means: state (1) as the headline bound with the two-line proof of Sec. 3; keep (95)–(97) as the flatness explanation, dropping the sample index i from them (they are function-level identities; i belongs only to (92) and the least-squares objective); and demote (98)–(107) to a remark scoped to what it is — a bound on the forcing response, not on the measured remainder.

9 Reproduction

`analysis/sim_envelope_witness.py`	simulation sweep → `docs/sim_env_witness_runs.csv`
`analysis/hw_envelope_witness.py`	hardware sweep → `docs/hw_env_noise_runs.csv`
`analysis/bound_final_figs.py`	caps of (1), both figures, the counts of Sec. 7
`analysis/de_measure.py`	the direct δe_ω measurement of Sec. 8

The residual in every script is the pipeline's own stage-1 call, `extract_piecewise(model='cosh', cosh_c2=None, ramp_gain=None)` — both constants and the onset free, the onset swept over the whole window with no auxiliary seed model, the full window, no calibration, and the reported RMSE exactly $\sqrt{\text{cost}/N_s}$ of the minimised objective.